Shane McGrath

Senior Software Engineer — Frontend & Design Systems
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Summary

Senior Software Engineer with 8+ years of experience architecting and delivering scalable, data-intensive web applications and data-visualisation tools. Experienced in the React/TypeScript ecosystem with deep experience in state management, front-end architecture & concepts. Proven track record of improving system reliability, upholding engineering standards, and driving product quality through automated testing. A collaborative teammate skilled at mentoring engineers and translating ambiguous requirements into robust end-to-end technical solutions.

Experience

Senior Software Engineer – Sustain.Life Aug. 2022 – Present
Enterprise carbon accounting platform built with React, TypeScript and GraphQL. *Remote, Ireland*

- Owned and developed complex, data-heavy frontend application using **React, TypeScript and GraphQL** within a large monorepo, driving architecture and implementation across complex product areas.
- Designed reusable frontend components and workflows with UI/UX, establishing consistent patterns through **Storybook** and reducing duplicated implementation across the product.
- Improved frontend reliability and development velocity by expanding automated test coverage, reducing daily manual QA effort by **45%**.
- Built end-to-end integrations with third-party providers including **Codat, Utility API and Arcadia**, supporting key product workflows.
- Worked across frontend and backend services in a distributed system using **microservices and event sourcing**.
- Owned production reliability and zero-downtime deployments, troubleshooting CI/CD pipelines, Kubernetes resources and critical incidents through on-call support.
- Interviewed new hires. Mentored engineers through pair programming, technical guidance and code reviews, helping establish maintainable engineering practices across the team.
- Translated ambiguous customer and product requirements into maintainable technical solutions with a focus on product outcomes.

Frontend Developer – Axial3D Aug. 2018 – July 2022
Medical technology startup delivering 3D printing and machine-learning workflows. *Belfast, NI*

- Built responsive frontend applications for displaying **medical image segmentations, 3D visualizations and annotations**, translating **Figma** designs into production workflows.
- Built interactive **3D medical visualization** workflows using **Three.js and WebGL**, enabling users to inspect and interact with machine-learning-generated medical imaging data in the browser.
- Reduced customer onboarding time by **35%** through usability improvements and redesign of the sign-up workflow.
- Rapidly delivered new product capabilities using customer analytics and feedback to validate product direction and prioritize engineering work.
- Designed and implemented **REST APIs** supporting customer workflows around medical imaging data in accordance with the **DICOM** standard.
- Implemented frontend automated testing using **Cypress and Jest**, improving regression coverage and confidence in production deployments.
- Applied software design patterns, SDLC practices, Git-based version control and testing strategies to build maintainable production software.
- Onboarded new engineers into the development environment, sharing technical knowledge, system context and development practices.
- Earned the **AWS Cloud Practitioner** certification, developing practical knowledge of cloud architecture, security and AWS services.

Design Engineer Intern – Product & 3D Modeling – PMD Solutions Apr. 2017 – June 2018
Contributed to product design for a sleep apnoea wearable device and companion application. *Cork, Ireland*

- Designed mechanical components in **SolidWorks** and developed rapid-prototyping workflows using 3D printing, including training team members in 3D printer usage.
- Conducted user research and collaborated with designers and developers on web and native mobile product development, contributing to roadmap decisions.

Education

University of Galway — Ireland
MSc Software Design and Development — *2 years*

University of Galway — Ireland
Bachelor of Engineering (Mechanical Engineering) — *4 years*

Technical Skills

Core Technologies: TypeScript, JavaScript, HTML, CSS, SQL
Frontend & Design Systems: React, Next.js, TanStack (Query, Table, Router), Vue, Tailwind CSS, Radix UI, Three.js, WebGL, Material UI, Figma
Testing: Jest, Vitest, Cypress, Selenium
Backend & APIs: Node.js, GraphQL, REST, tRPC
Cloud & Infrastructure: AWS, Docker, Kubernetes, CI/CD
Tools & AI: Git, Bun, Storybook, Vite, Webpack, Postman, Claude Code (AI Coding Agents), Sentry